NANYANG TECHNOLOGICAL UNIVERSITY

SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2 MH1301 Discrete Mathematics Tutorial 11

For the tutorial in Week 12 on 11 April, let us discuss the following questions.

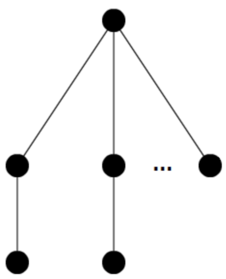
Questions 1–3. Assume that $n \geq 7$,

1. How many (non-isomorphic) trees with n vertices are there in which one vertex has degree $n-1$?
2. How many (non-isomorphic) trees with n vertices are there in which one vertex has degree $n-2$?
3. How many (non-isomorphic) trees with n vertices are there in which one vertex has degree $n-3$?

Solution:

1. A tree with n vertices has $n-1$ edges, so if a vertex has degree $n-1$, the graph must be $K_{1,n-1}$ (also called a star). That is, there is only 1 (non-isomorphic) tree of this type.
2. If one vertex v has degree $n-2$ then we have a star $K_{1,n-2}$ plus one more edge and one more vertex v_n. Clearly v_n is not connected to the vertex v with degree $n-2$, because then the degree of that vertex would be $n-1$. Hence v_n is connected to some other vertex. Since they all look the same, there is only 1 (non-isomorphic) tree of this type.
3. Now suppose a vertex v has degree $n-3$. First note that no other vertex has degree $n-3$ as well, since they would be connected to $2(n-3)-1$ edges together (they might share 1 edge). Using the fact that $n \geq 7$, we see that the number of edges will be at least $2n-7 \geq n$, thus, the graph would not be a tree.

   Hence there is exactly one vertex with degree $n-3$, and we can start with a $K_{1,n-3}$. There are now 2 other vertices and 2 edges. These can either be connected to two of the other $(n-3)$ vertices (A), both to one of the other vertices (B), or as a path to one of the other vertices (C). Hence there are 3 (non-isomorphic) graphs of this type.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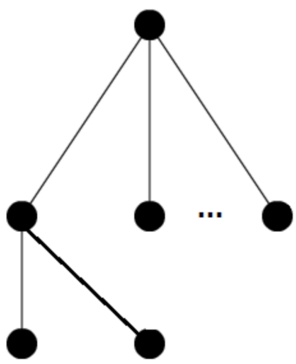
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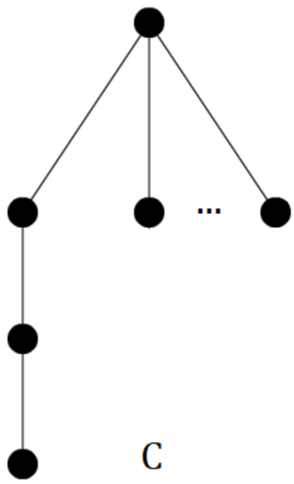
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Question 4. (Ex 11.1.8). Prove that a simple graph is a tree if and only if it is connected, but the deletion of any of its edges produces a graph that is not connected.

Solution: We need to prove this statement in two directions:

($\Rightarrow$) Suppose we are given a simple graph that is a tree, and we delete an edge, say (u, v), from the tree. We want to show that the resulting graph is disconnected.

We argue that there will be no path between u and v in the resulting graph. Otherwise assume there is still a path P that connects u and v. Then combining P and (u, v) will give a circuit in the original tree, which contradicts the tree definition.

($\Leftarrow$) For this direction, we suppose that we are given a simple graph with the property that it is connected, but the deletion of any of its edges produces a graph that is not connected. We use a proof by contradiction.

Suppose not. That is, suppose to the contrary, that the graph is not a tree. Then it contains a circuit $C = (v_1, v_2, \ldots, v_n, v_1)$. Now consider a subgraph G' defined from G by deleting an edge, say (v_1, v_2), from circuit C. We claim that G' is still connected. This is because for any path P in G between any two vertices x and y, if (v_1, v_2) is not in P, then x and y are connected in G'; if $(v_1, v_2) \in P$, x and y are still connected through P and the rest of circuit $(v_2, v_3, ..., v_n, v_1)$. Hence, G' is still connected, which contradicts to the assumption.

Question 5. (Ex 11.1.10). Which complete bipartite graphs $K_{m,n}$, where m and n are positive integers, are trees?

Solution: Suppose both m and n are at least 2. Then there is a simple circuit of at least 4 in $K_{m,n}$, and so $K_{m,n}$ is not a tree. So we need either $m = 1$ or $n = 1$ (or both).

On the other hand, when $m = 1$, we can easily see that $K_{1,n}$ is a tree. Likewise, $K_{m,1}$ is also a tree. Therefore, $K_{m,n}$ is a tree if and only if $m = 1$ or $n = 1$.

Question 6. (Ex. 11.1.11, 11.1.13).

(a) How many edges does a tree with 10000 vertices have?

(b) How many edges does a full binary tree with 10000 internal vertices have?

Remark: Full binary trees have odd number of vertices and so a full binary tree cannot have 10000 vertices.

Solution:

(a) Recall that a tree with n vertices has $n-1$ edges. Therefore a full binary tree with 10000 vertices has 9999 edges.

(b) Since this is a full binary tree, every internal vertex v has two children. Therefore, there are two edges connecting v to its two children. Since in a rooted tree, every edge is incident with a parent and a child, the number of edges is the same as the number of children in the tree. Since there are 10000 internal vertices, we conclude that there are $2 \times 10000 = 20000$ edges.

Question 7. (Ex 11.1.23). How many edges are there in a forest of t trees containing a total of n vertices?

Solution: Answer is $n - t$.

Suppose the forest consists of trees $T_1, T_2, \dots, T_t$ with vertices $n_1, n_2, \dots, n_t$, respectively, with the sum of vertices

$$n_1 + n_2 + \cdots + n_t = n.$$

For each tree T_i, with $1 \leq i \leq t$, there are $n_i - 1$ edges. So the total number of edges in the forest is

$$(n_1 - 1) + (n_2 - 1) + \cdots + (n_t - 1) = n - t.$$

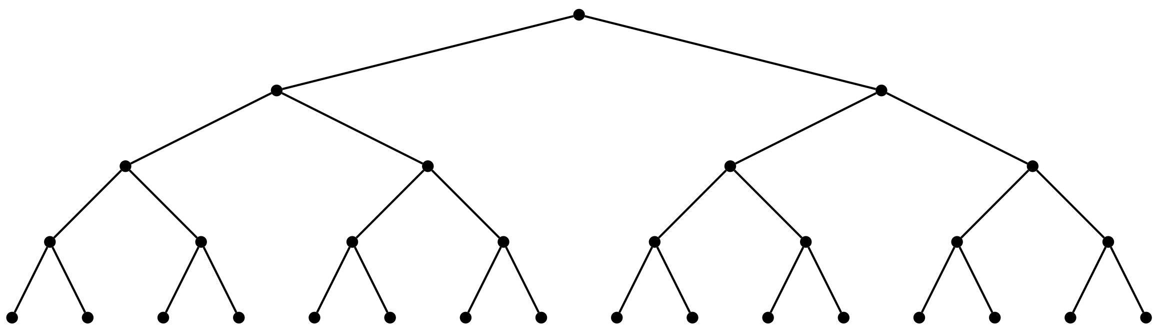
Question 8. (Ex 11.1.19, 11.1.20). A **complete binary tree** is a full binary tree in which every leaf is at the same level.

(a) Construct a complete binary tree of height 4.

(b) How many vertices and how many leaves does a complete binary tree of height h have?

Solution:

(a) Here is a complete binary tree of height 4.



(b) A complete binary tree has 2^k vertices at level k for $1 \leq k \leq h$. So the total number of vertices is

$$2^0 + 2^1 + \cdots + 2^h = 2^{h+1} - 1.$$

The number of leaves is equal to the number of vertices at level h, which is 2^h.

Answer.

1) 1.
2) 1.
3) 3.
5) $m = 1$ or $n = 1$.
6) 9999; 20000.
7) $n - t$.
8) 2^h.